

AYA TLIJANI

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PROFILE

Engineering student in Computer Science, specializing in **Data Science and Artificial Intelligence**, with experience in **data analysis, machine learning, and software development**. Seeking opportunities in **data science, artificial intelligence, and software development**.

PROFESSIONAL EXPERIENCE

University of East London – Research Laboratory

Jul 2026 - Sep 2026

Junior Research Assistant (Remote)

Remote

Working on applying artificial intelligence to project management and resource-constrained scheduling optimization.

- Developing a solution for the **Resource-Constrained Project Scheduling Problem (RCPSP)**, incorporating **PSPLIB instances**, scheduling environment simulation, and **graph-based project representation**.
- Implementing a **Graph Isomorphism Network (GIN)** with **Proximal Policy Optimization (PPO)** using **PyTorch** and **PyTorch Geometric** to learn scheduling policies from graph-structured project data and optimize **project makespan**.

AMSYS Consulting

Aug 2026 - Sep 2026

Artificial Intelligence Intern

- Contributed to the design and development of **AI pipelines** as part of a **confidential project**.
- Designed and evaluated **AI workflows** for intelligent analysis solutions.

ESPRIM

Jul 2026 - Aug 2026

Data science intern

Contributed to the design of a five-competency framework and associated evaluation grids within the industrial project I5-Assist.

- Developed a **pipeline** for extraction and **feature engineering**, transforming learning data into measurable competency indicators.
- Applied **data analysis** and **NLP techniques** to extract competency indicators from structured and unstructured educational data.

GCT Gafsa

Jul 2025 - Aug 2025

Introductory Internship

IT support, exploration of the company's network infrastructure, and participation in the development of an e-commerce platform with PrestaShop.

PROJETS

Fraudex AI - Bank Fraud Detection with Deep Learning

- Analyzed and explored a **Kaggle transaction dataset** using **Python**, simulated a **real-time transaction stream**, and implemented a process to detect **fraudulent patterns**.
- Designed and deployed a **deep learning pipeline** in **TensorFlow** combining an **autoencoder**, **attention mechanisms**, and **transformer architectures** to assign fraud scores to transactions and assess their potential fraud risk.
- Evaluated and compared **detection models** using **performance metrics** to analyze their accuracy in identifying fraudulent transactions.

Brain Tumor Detection – Medical Image Classification

- Implemented a **convolutional neural network (CNN)** in **Python** to classify brain tumors from **MRI images**.
- Applied **data augmentation** techniques and evaluated model performance using **accuracy, precision, and recall** metrics, with analysis of the results.

ArtisanConnect – E-commerce Platform for Artisans

- Developed a **desktop (JavaFX)** and **web (Symfony/PHP)** e-commerce application integrating a product catalog, shopping cart, and user management.

EDUCATION

ESPRIT Monastir (ESPRIM)

Sep 2024 - Present

Engineering Degree in Computer Science, specializing in Data Science and Artificial Intelligence

Monastir

Preparatory Institute for Engineering Studies of Monastir (IPEIM)

Sep 2021 - Jun 2024

Preparatory Institute for Engineering Studies

Monastir

KEY SKILLS

- **Programming Languages:** Python, Java, PHP, SQL, C, C#
- **Web Development:** HTML, CSS, JavaScript
- **Frameworks:** Symfony, JavaFX, React, Angular
- **Tools:** Git, Jupyter Notebook, Power BI, Databricks

CERTIFICATIONS

- NVIDIA Deep Learning Institute
- Certificat en fondamentaux de l'apprentissage profond

LANGUAGE SKILLS

- Arabic(Native language)
- English(C1 level)
- French(B2 level)
- Spanish(Basic level)